University of Verona

A.Y. 2025-26

Machine Learning & Deep Learning
Deep Learning

Projects and Exam

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Final Evaluation

Project

1) Code shared using GitHub

2) Written Report (to be evaluated)

3) Oral Presentation

Theory

A few questions about the theory to be asked during the project presentation

Project – Rules

- The project can be carried out in groups of 2 people max
- 3 persons are exceptionally acceptable if duly justified, but 2 people max is the norm
- There is the possibility to present a public *pitch* of 5 min (in class) describing the project (no more than 5 slides), and we'll give you feedback and suggestions or contact directly the teachers. – take this opportunity sending us an email by April 2026
- Alternatively, just contact us if you need clarification or you have doubts on whatever aspect of your project: we will meet or reply by email to give you feedback
- Possibility to perform projects in collaboration with other courses, subject to agreement between the teachers
- Possibility of Master theses

Project – Rules

- When the project is concluded, each group should present:
  - A written technical report, to be evaluated by us
  - Project code, shared using GitHub
  - An oral presentation of about 8-10 min, to be done by all authors by preparing a PowerPoint presentation
  - After the presentation, a few questions related to the theory will be asked
- The official dates of the exams will be fixed when the booking period will be open, **BUT** they are ONLY MANDATORY for the exam registration and the delivery of the report to us
- The ACTUAL EXAM, that is, the project discussion + theory questions, will be held in a date to be fixed, which is typically about a week after the official dates.

Typical project structure

- The typical project should be structured as follows:
  - **Problem or task definition**: classification, regression, clustering, etc.., and associated application
  - **Dataset selection** and description
  - **Feature extraction** from data and related analysis (mainly in case of ML). A discussion about features and representations is also welcome in case of DL project
  - **Model/Methodology/Architecture** to be utilized
  - Evaluation of the **Results** and analysis

Dataset Selection

- Strongly constrained to the tackled problem

- Example datasets:
  - [LEGO Bricks]
  - [CelebA] : celebrity faces
  - Other datasets and project ideas on Kaggle.com or https://paperswithcode.com/datasets

- Or you can google the task your want to address + the word «datasets», *e.g.*, «face recognition datasets»

Model/Method/Architecture

- Utilize deep models seen during the lectures to solve the tackled task

- Make comparisons, at least 5 variants:
    - Same model with different (hyper-)parameters
    - Different models, *e.g.*, CNNs, transformers, etc.

Result evaluation and analysis

- Every single variant should be endowed of:
  - Quantitative results:
    - Confusion matrix
    - Accuracy figures (e.g., IoU, etc)
    - Recall/Precision figures (whenever possible)
  - Qualitative results:
    - Images representing input, output/predictions, ground-truth

How to write a technical report

- No fixed number of pages, not too short, not too long, if you want numbers let's say within 8 and 20 pages, but it's not mandatory

- **COVER PAGE:** Project Title, Master degree, Course, Academic Year, Author(s)
- Index (structure of the report)
- Subdivision in the following **Sections** (next slide)

- The following suggested subdivision in Sections is indicative of the logical structure of the report, but not necessarily the only possible one

How to write a technical report

- Section MOTIVATION AND RATIONALE
  - Place the proposed project in context, possibly identifying a Research Theme(s) of interest. What is the problem addressed by your project? Why is it significant?
- Section STATE OF THE ART [optional, but at least a glance]
  - Describe the state of the art relevant for the project
  - What results or techniques do you plan to exploit? Which are the weak points of the SoA methods, which ones need to be improved? Why? How?
- Section OBJECTIVES
  - Please, clearly state the general and specific objectives of your project

How to write a technical report

- Section METHODOLOGY
  - Discuss method(s), algorithms, datasets, analytical and computational tools that will be necessary to pursue the project objectives
- Section EXPERIMENTS & RESULTS
  - Describe the evaluation protocol, conditions, datasets used and metrics to measure the performance (accuracy, confusion matrix, recall & precision, etc.)
- Section CONCLUSIONS
  - Summarize the project (goals, methods, results) and sketch possible future works
- Section BIBLIOGRAPHY or REFERENCES
  - Only list the bibliographic references that are strictly relevant for describing the research project. All references should correspond to a citation in the text

How to write a technical report: Chat GPT, etc.

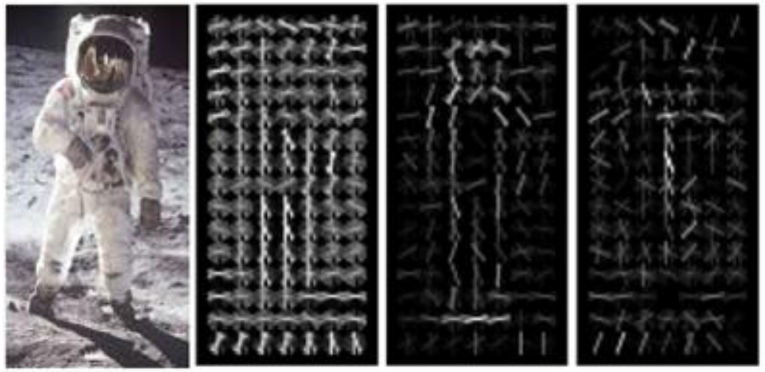
- ChatGPT is not allowed to write the report and no section should be written with automatic tools (ChatGPT, Gemini, …)
- Every report will be checked with detectors in order to understand how much of your text is written by you
- ChatGPT is allowed to rewrite the text or to check for typos. In that case, we suggest to use **Grammarly** that is not detected by our detectors
- **When you'll send the report, if the case, please specify every part in which you used ChatGPT or similar systems to revise the text**

Project example

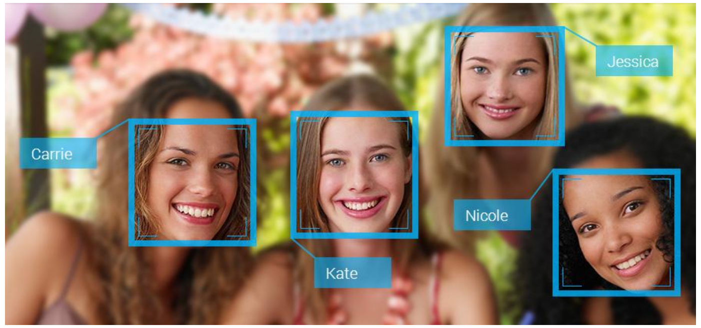
- Hair color classification of every celebrity present in the CelebA dataset
- Features: extracted by a pre-trained ConvNet
- Classifiers:
  - CNN
  - Transformers
  - AE/VAE/GAN
  - ViT
  - .....

Possible further themes

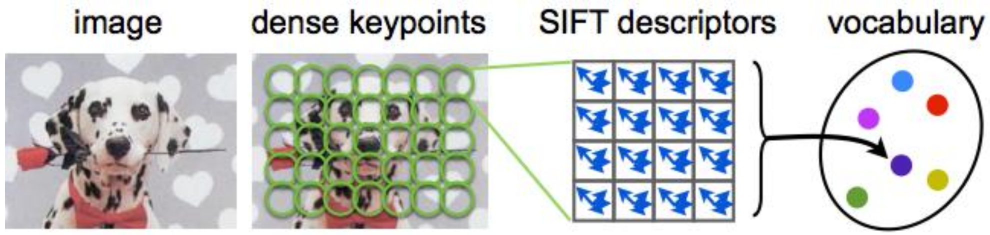
- Pedestrian detection (HOG features)
    - Other object detection problems: cars, bikes, …



- Face detection (and recognition)

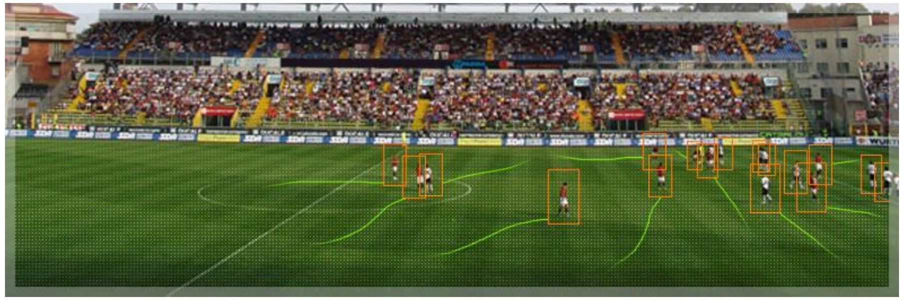


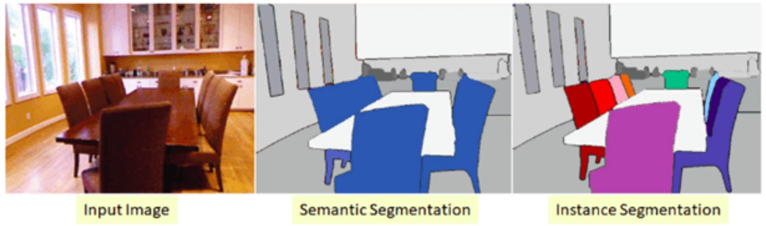
- SIFT features: classification, recognition, detection



Possible further themes

- Sequence-like problems: estimate motion (*e.g.*, optical flow) and classification
- Detection and tracking (keypoints)
- Image (semantic) segmentation
- Music classification, or other types of data, not necessarily images
- Human activity recognition
- … your proposals





Development environments

SciPy

NumPy

matplotlib

IP[y]: IPython Interactive Computing

pandas

scikit learn